Question 1: differentiation

(a) Differentiate $\dfrac{x^{1/2}-2}{x^{1/2}+1}$.

Let $u = x^{1/2} - 2$ and $v = x^{1/2} + 1$, so that $u' = v' = \frac{1}{2}x^{-1/2}$. Using the quotient rule, for $x > 0$,

$$\begin{aligned}\left(\frac{u}{v}\right)' &= \frac{vu' - v'u}{v^2} \\ &= \frac{(x^{1/2}+1)\,\frac{1}{2}x^{-1/2} - (x^{1/2}-2)\,\frac{1}{2}x^{-1/2}}{(x^{1/2}+1)^2} \\ &= \frac{\left[(x^{1/2}+1) - (x^{1/2}-2)\right]\frac{1}{2}x^{-1/2}}{(x^{1/2}+1)^2} \\ &= \frac{3}{2\,x^{1/2}\,(x^{1/2}+1)^2}.\end{aligned}$$

(b) Differentiate $x^{(2x)}$.

Note: power rule or chain rule can't be used yet, since $x = e^{\ln x}$ for $x > 0$, we have $(x)^k = (e^{\ln x})^k$, and so $x^{2x} = e^{2x \ln x}$. We also need $\dfrac{d}{dx}e^{f(x)} = f'(x)e^{f(x)}$.

If $f(x)$ is in the exponent, first lets get $f'(x)$, then build $f'(x)\exp[f(x)]$

$$\frac{d}{dx}[2x \ln x] = (2 + 2\ln x) \tag{1}$$

$$\Rightarrow \frac{d}{dx}e^{2x \ln x} = (2 + 2\ln x)e^{2x \ln x} \tag{2}$$

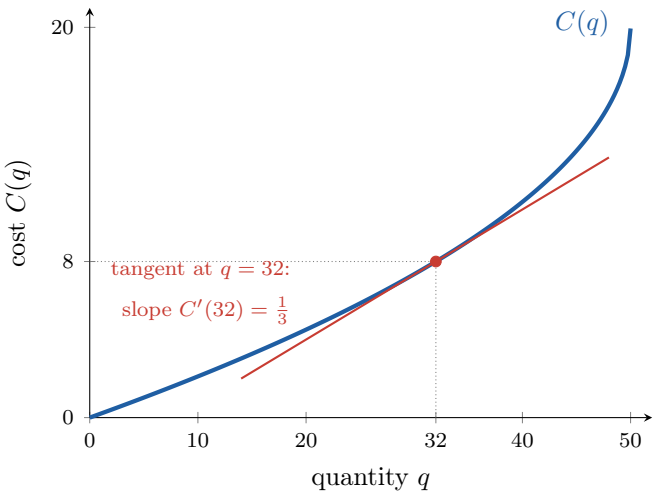
$$= 2(1 + \ln x)x^{2x} \tag{3}$$

Question 2: marginal cost

Suppose that the cost function is given by

$$C(q) = 20 - 4\left(25 - \frac{q}{2}\right)^{1/2}$$

where q is quantity produced. Find the marginal cost when $q = 32$. Hint: Think of the marginal cost as the change in the value of $C(q)$ for a very small change in q.



Marginal cost is the slope of the cost function, imagining costs were to change due to a a very small change in quantity produced. Particularly useful concept for optimality conditions which dictate at the optimum choice there should be no possible profitable deviation, no matter how small. This will usually imply some marginal benefit mins marginal cost equals zero, or more commonly, MB=MC.

$$C'(q) = (-4)(1/2)(25 - q/2)^{-1/2}(-1/2) \tag{4}$$

$$= (25 - q/2)^{-1/2} \tag{5}$$

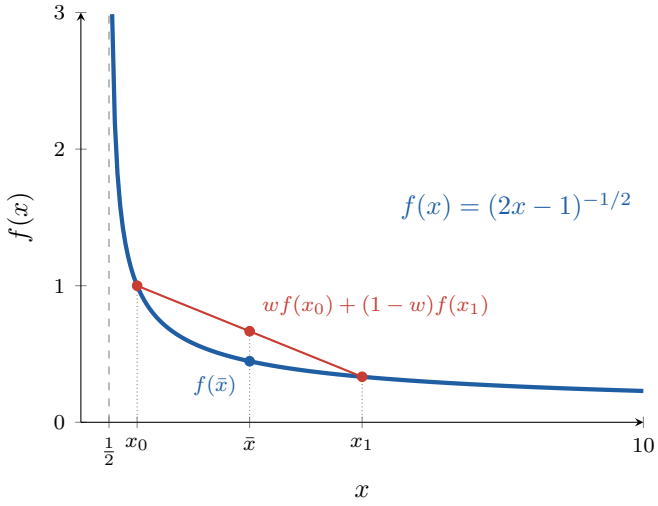
$$C'(32) = (25 - 16)^{-1/2} \tag{6}$$

$$= 1/3 \tag{7}$$

Question 3: concave or convex

Is the function $f(x) = (2x-1)^{-1/2}$ concave or convex on its domain?

For a function of a single input and a single output, checking covexity/concavity is relatively straightforward (a bit more involved if $z = f(x, y)$, will see in later weeks.



$$\text{level: } f(x) = (2x-1)^{-1/2} > 0 \quad \text{defined for } x > 1/2 \tag{8}$$

$$\text{slope: } f'(x) = -(1/2)(2)(2x-1)^{-3/2} = -(2x-1)^{-3/2} < 0 \tag{9}$$

$$\text{curvature: } f''(x) = (3/2)(2)(2x-1)^{-5/2} = 3(2x-1)^{-5/2} > 0 \tag{10}$$

WE can deduce the curve is downward sloping over its full support, however the slope becomes shallower as x increases. This satisfies strict convexity:

Definition: Convexity. For some weight w between 0 and 1 (100 percent), for any two distinct points x_0, x_1:

$$w \cdot f(x_0) + (1-w) \cdot f(x_1) > f(w \cdot x_0 + (1-w) \cdot x_1)$$

$$Average\ f > f(average)$$

the average of the function (evaluated at two points) is strictly above the function evaluated at an average of the two input points. Drawn in red above.

Question 4: elasticity

(a) Prove that if $F(x) = \dfrac{f(x)}{g(x)}$, then $\dfrac{F'(x)}{F(x)} = \dfrac{f'(x)}{f(x)} - \dfrac{g'(x)}{g(x)}$.

Assume f and g are differentiable with $f(x) \neq 0$ and $g(x) \neq 0$. By the quotient rule,

$$F'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}.$$

Dividing by $F(x) = f(x)/g(x)$, is the same as multiplying by $g(x)/f(x)$

$$\begin{aligned}\frac{F'(x)}{F(x)} &= \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2} \cdot \frac{g(x)}{f(x)} \\ &= \frac{f'(x)g(x) - f(x)g'(x)}{f(x)\,g(x)} \\ &= \frac{f'(x)}{f(x)} - \frac{g'(x)}{g(x)}.\end{aligned}$$

(b) Using your result from part (a), show that the elasticity $\varepsilon_{F,x} = \varepsilon_{f,x} - \varepsilon_{g,x}$.

Definition. The elasticity of a function $h(x)$ with respect to x is the percentage change in $h(x)$ per one percent change in x:

$$\varepsilon_{h,x} = \frac{\%\text{ in } h}{\%\text{ in } x} = \frac{\frac{d\ln h}{dx}}{\frac{d\ln x}{dx}} = \frac{h'(x)/h(x)}{1/x} = x \cdot \frac{h'(x)}{h(x)}$$

Applying the definition: the three elasticities in the question are

$$\varepsilon_{F,x} = x\,\frac{F'(x)}{F(x)}, \qquad \varepsilon_{f,x} = x\,\frac{f'(x)}{f(x)}, \qquad \varepsilon_{g,x} = x\,\frac{g'(x)}{g(x)}.$$

Part (a) showed that

$$\frac{F'(x)}{F(x)} = \frac{f'(x)}{f(x)} - \frac{g'(x)}{g(x)}.$$

Multiplying both sides by x,

$$x\,\frac{F'(x)}{F(x)} = x\,\frac{f'(x)}{f(x)} - x\,\frac{g'(x)}{g(x)},$$

and each term is an elasticity, so

$$\varepsilon_{F,x} = \varepsilon_{f,x} - \varepsilon_{g,x}.$$

Interpretation. for each percent change in x, f responds by $(\varepsilon_{f,x})\%$ and g by $(\varepsilon_{g,x})\%$, then the ratio $F = f/g$ changes by approximately $(\varepsilon_{f,x} - \varepsilon_{g,x})\%$: the percentage change in a ratio is the percentage change in the numerator minus the percentage change in the denominator.

Question 5: income elasticity of demand

Demand $D(I) = AI^{1.23}$. How does demand change for a percentage change in income?

Fairly straight forward:

$$\varepsilon(D, I) = I \cdot \frac{D'(I)}{D(I)} = \frac{(1.23)AI^{1.23-1}I}{AI^{1.23}} = 1.23 \tag{11}$$

for a 1 percent increase in income will lead to a 1.23 percent increase in demand, approximately.

Interpretation: If elasticity is greater than 1 we call demand "elastic", quantity demanded responds more than one-for-one on a proportional basis to a change income. If elasticity is less than 1 we call demand "inelastic".